Try This Eye Trick: See Colors That Are Not There

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Here is a fun trick. Stare at a colorful picture for 30 to 60 seconds. Then look at a plain white wall or paper. You will see a **ghost-like** copy of the picture in different colors! This is called an **afterimage**.

Your eyes see color with special cells called **cone cells**. There are three kinds, often called red, green and blue. When you look at something red, your red cones tell your **brain**, "That is red!"

When red, green and blue light mix, they make white. So when you look at something white, all three kinds of cones work together.

But **cone cells** can get tired. If you stare at a red circle for a long time, your red cones take a break. Then, when you look at white, only your green and blue cones work. So you see a blue-green circle!

After a few seconds, your red cones wake up again, and the **afterimage** goes away. Try it with different colors.

Word Treasure

afterimage -- A ghost-like picture you see after staring at something and looking away.

cone cells -- Cells in your eyes that help you see colors.

brain -- The organ in your head that helps you think and understand what you see.

ghost-like -- Pale and faint, like a ghost.

Background you need

Your eyes have special cells that sense light and color.

Red, green and blue light can mix together to make white light.

When cells in your eyes work hard for a long time, they can get tired.

Break it down

WHO: You and the cone cells in your eyes
WHAT: Seeing an afterimage in new colors after staring at a picture
WHERE: Looking at a plain white wall or paper
WHY: Tired cone cells stop working for a moment, so you see different colors

Why it matters

It is a fun way to learn how your own eyes and brain work together.

Test what you remember

1. How long should you stare at the colorful picture?

- (A) 10 minutes
- (B) 1 second
- (C) All day
- (D) 30 to 60 seconds

2. What is an afterimage?

- (A) A painting
- (B) A photo on a phone
- (C) A ghost-like copy of a picture in different colors
- (D) A shadow

3. What cells help your eyes see color?

- (A) Bone cells
- (B) Skin cells
- (C) Hair cells
- (D) Cone cells

4. What do red, green and blue light make when mixed?

- (A) White
- (B) Black
- (C) Orange
- (D) Gray

5. After staring at a red circle, what color do you see?

- (A) Purple
- (B) Yellow
- (C) Red
- (D) Blue-green

6. Why does the afterimage go away?

- (A) You fall asleep
- (B) The light turns off
- (C) Your red cones wake up again
- (D) The paper changes color

Answer key (try the quiz first!): 1: D 2: C 3: D 4: A 5: D 6: C

Questions to think about

MY THOUGHTS (at least 20 words):